

EcoQuest – An Open World Sustainability Game

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ABSTRACT

EcoQuest is an immersive virtual reality experience designed to teach players the impact of environmental decision making through interactive gameplay. Set on a fragile island ecosystem, players harvest resources and manage an inventory while their actions dynamically alter environmental health and endgame outcomes. Unlike traditional survival games, EcoQuest ties player behavior to persistent ecological change: deforestation reduces ground cover, wildlife disappears, and the island becomes barren if resources are mismanaged. EcoQuest demonstrates how VR can promote environmental awareness by combining embodied interaction with real-time ecological feedback.

Index Terms: Virtual Reality, Sustainability, Environmental Awareness, Serious Games, Interaction Design.

1 INTRODUCTION

Environmental education games often rely on 2D interfaces, textual explanations, or abstract simulations that limit a player's sense of presence and emotional connection to ecological processes. Virtual Reality (VR) offers an alternative medium in which consequences can be experienced in an embodied and immersive environment [8]. This makes VR a powerful medium for teaching sustainability through interactive feedback and environmental storytelling [1, 5]. VR supports embodied interaction, agency, and emotional salience, making environmental consequences feel personally meaningful [7, 8].

EcoQuest is a VR survival style experience that reframes resource collection as an ecological responsibility rather than an unlimited mechanic [2]. Players interact with trees, terrain, and wildlife, and these actions immediately alter the world state [5]. As vegetation disappears, the terrain shifts from lush grass to barren soil, animals vanish, and the island progressively "dies." The game includes multiple endings based on player choices, providing a clear connection between individual behavior and environmental impact.

Our goal is to create a VR system where the environment responds systematically to the player's decisions, enabling users to learn about sustainability through embodied interaction rather than passive instruction. By integrating real-time terrain modification, timed challenges, inventory management, and dynamic feedback systems, EcoQuest demonstrates the potential of VR as a tool for ecological awareness and interactive education.

2 RELATED WORK

Prior research has explored the use of Virtual Reality as a tool for fostering environmental awareness, empathy, and sustainability oriented learning [6]. Immersive VR experiences have been shown to increase users' sense of presence and emotional engagement compared to traditional 2D media, which can support reflection on environmental consequences and pro-environmental attitudes [8, 4].

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Several studies have used VR to simulate ecological systems such as deforestation, climate change, and resource depletion, often through scripted or guided experiences designed to evoke empathy or illustrate cause effect relationships [2, 3]. More recent work has emphasized the role of interactivity and agency in environmental serious games, suggesting that active decision making and feedback driven outcomes can enhance learning and awareness [2].

Unlike many prior VR sustainability experiences that rely on fixed narratives or single outcome scenarios, EcoQuest adopts an open ended survival gameplay structure. Players are given continuous agency over resource use, with persistent environmental degradation and multiple failure states emerging directly from their actions [1]. This design allows ecological consequences to be experienced dynamically rather than presented as scripted events, positioning EcoQuest as a player driven, outcome sensitive VR system for sustainability education.

3 USER EXPERIMENT

To evaluate whether EcoQuest can increase players' awareness of ecological consequences specifically deforestation and unsustainable resource extraction we designed the following pilot user study.

3.1 Participants

Five participants (ages 19–27), from a group of volunteers, played the VR game for approximately 10 minutes each. All participants had prior experience with VR or 3D games but had no prior exposure to the EcoQuest environment. None reported significant vision or motor impairments that might affect gameplay.

3.2 Apparatus

The experiment used a Meta Quest 3S headset and a Unity engine build of EcoQuest (terrain, grass, pine trees, apple/pear trees, bushes, rocks; cutting and inventory system; hunger mechanic; island fast forward sequence), with a desktop PC for observer logging and Unity analytics.



(a) Initial game state.

(b) Barren game state.

Figure 1: Representative EcoQuest gameplay states showing the transition from an initially lush island environment to a barren landscape following sustained overharvesting, highlighting the ecological consequences of player actions.

3.3 Task

Participants were instructed to explore the island freely while completing the basic available quest: locate apple/pear trees, cut trees to gather food items, cut trees to collect fruit items, and try to survive for the duration of the 10 minute demo.

They were not told that cutting trees would trigger larger ecological consequences. After cutting eight trees, the system fast forwards the world, showing a barren island with a message warning against reckless resource extraction.

Table 1: Key quantitative results from the EcoQuest pilot study (N=5).

Metric	Result
Quest Completion Rate	100%
Average Awareness Score	4.3 / 5
Average Enjoyment Rating	4.1 / 5
Hunger Deaths	2 / 5 players
Players Reaching Fast Forward Ending	3 / 5 players

EcoQuest uses a **fixed timebased hunger rule** with an inventory requirement: every 50 seconds, the game automatically consumes one fruit (apple or pear) from the player's inventory; if no fruit is available at a 50 second check, the player dies immediately as shown in Fig. 1; and the fast forward ending triggers after the player cuts eight trees (independent of the hunger rule). The study duration was 10 minutes, but sessions can end earlier due to either hunger-death or the eight tree fast forward event.

3.4 Measures

Based on our evaluation plan (awareness score, enjoyment rating, quest completion rate), we collected both quantitative and qualitative measures. Quantitative measures included the **Environmental Awareness Score** (1–5 scale): “To what extent did the game make you aware of ecological impact?”, the **Enjoyment Rating** (1–5 scale), **Quest Completion Rate**, **Number of Trees Cut Before Noticing Impact**, and **Survival Duration** (until hunger death or tree quota fast forward event). For each participant, the Environmental Awareness Score was computed as the mean of the three 1–5 Likert items. We then report the overall awareness result as the mean of participants' composite scores.

We defined project success *a priori* as a **mean Environmental Awareness Score** $\geq 4.0/5$. On a 5 point Likert scale, values above 4 correspond to strong agreement, and are commonly used in formative HCI/educational game evaluations to distinguish neutral or mildly positive responses ($\approx 3-4$) from strongly positive perceived learning impact (> 4). The observed mean of **4.3/5** therefore indicates consistently high awareness across participants and satisfies the predefined success criterion.

Post study open ended questions included: “What did you understand about ecological consequences from this experience?”, “What aspects helped or hindered your understanding?”, and “How did the barren island sequence affect your thinking?”

4 PRELIMINARY STUDY RESULTS

4.1 Quantitative and Qualitative Findings

Table 1 summarizes the key quantitative results from the pilot study.

All participants successfully completed the core quest and reached either the hunger-death event or the ecological fast forward ending. The average environmental awareness score was 4.3, meeting the project's success criteria, and enjoyment remained high at 4.1. Two participants died from hunger before reaching the tree cutting threshold, indicating they had not collected enough fruit early in the game. The remaining three participants reached the barren island fast forward event, which strongly communicated the ecological consequences of overharvesting. Overall, the quantitative results show that players understood the resource survival connection and were receptive to the game's ecological messaging.

The fast forward barren island reveal was the strongest trigger of ecological reflection, described as “shocking” and making players' actions “actually matter.” Participants noted that it showed how individual choices accumulate into long term environmental consequences. The hunger mechanic reinforced this by creating

trade-offs between immediate survival and long-term environmental health, encouraging tree cutting while highlighting the risks of overharvesting and changing assumptions about resource abundance.

5 CONCLUSION

This pilot study suggests that EcoQuest, a VR open world sustainability experience can support increased ecological awareness by linking player actions (tree cutting) to visible long-term environmental degradation. The observed awareness scores (4.3/5) and qualitative reactions indicates fast forward consequences, resource scarcity, and the visual transformation from a lush island to a barren landscape may enhance environmental learning and reflection.

Key findings indicate that VR immersion appeared to support strong engagement and reflection around environmental impact, consistent with patterns reported in prior VR user studies [7]. The hunger mechanic successfully established tension between survival needs and sustainability considerations, and all players completed the core quest, meeting our milestone criteria.

At the same time, the study has several limitations. The small sample size (N=5) limits statistical generalizability, and the short gameplay duration may not capture long-term engagement or learning retention. Furthermore, the current prototype focuses on a single ecological consequence (deforestation), whereas real ecosystems are highly interconnected.

6 FUTURE WORK

Building on our evaluation and player feedback, future work will explore dynamic ecosystem feedback (e.g., wildlife loss, soil erosion, reduced rainfall), static versus dynamic world comparisons to assess learning effects, longer-term studies of awareness retention beyond the VR session, and tree planting and regeneration mechanics to examine restorative behavior. Together, these directions will help determine how VR sustainability experiences can move from short-term awareness to lasting changes in environmental attitudes and decision-making.

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